

Machine Learning Analysis Report

Project 3 — Applied Machine Learning (Machine Learning Foundations)

1. Overview

This project applies supervised machine learning to **IoT-based fire detection**, with the goal of reducing false alarms while maintaining high recall on real fire events. The dataset used is the **Smoke Detection Dataset** (Kaggle, `deepcontractor/smoke-detection-dataset`), a real, non-synthetic, publicly available multi-sensor recording from an IoT fire-detection prototype ([deepcontractor, 2022](See References)). The task is framed as **supervised binary classification** of the `Fire Alarm` target (0 = no fire, 1 = fire) from sensor readings. We compare a transparent, hand-tuned threshold rule against three supervised classifiers (Random Forest, Gradient Boosting, Logistic Regression) and one unsupervised anomaly detector (Isolation Forest).

The notebook implements the full workflow — problem definition, data loading and inspection, preprocessing, model training, evaluation, and a written summary — and additionally contains a self-contained **leakage audit** ("A second look" section) that re-evaluates every model under a leak-proof cross-session protocol.

2. Dataset Description

The Smoke Detection Dataset contains **62,630 rows** and **15 columns** (13 sensor features, a Unix timestamp `UTC`, a sample-count column `CNT`, and the binary target `Fire Alarm`). There are **no missing values**, and all features are numeric (float64 or int64), so no imputation or categorical encoding is required for the raw features.

The sensor features are:

- **Temperature [C]** — ambient temperature in degrees Celsius.
- **Humidity [%]** — relative humidity.
- **TVOC [ppb]** — total volatile organic compounds, in parts per billion.
- **eCO2 [ppm]** — equivalent CO₂, in parts per million.
- **Raw H2 and Raw Ethanol** — raw MOS sensor outputs for hydrogen and ethanol.
- **Pressure [hPa]** — atmospheric pressure.
- **PM1.0, PM2.5** — particulate-matter mass concentrations.
- **NC0.5, NC1.0, NC2.5** — number concentrations of particles in the named size bins.

The target distribution is **71.5% alarm (`Fire Alarm` = 1) vs. 28.5% no-alarm**, a moderate class imbalance that motivates the use of precision/recall/F1 rather than accuracy alone.

A property of this dataset that proved decisive for the evaluation is its **temporal structure**: as inspected in the notebook, 62,625 of the 62,629 row-to-row time gaps are exactly one second, and only four gaps exceed sixty seconds. The data is therefore a small number of continuous recording *sessions* from a single device, not 62,630 independent samples.

3. Modeling Approach

3.1 Preprocessing

Preprocessing follows the principle that every transformation fitted on the training data must not see the test data (Hastie, Tibshirani, & Friedman, 2009). Concretely:

1. **Feature engineering** — a cyclical time-of-day feature was derived from UTC using sine and cosine encoding (`hour_sin`, `hour_cos`), so that midnight (hour 0) and 11 p.m. (hour 23) are close in feature space rather than numerically distant (Kuhn & Johnson, 2013). The raw UTC timestamp and the CNT sample-count column were then dropped: the former is represented by the engineered features, and the latter is firmware metadata, not a physical signal.
2. **Scaling** — features were standardized with `StandardScaler` fit on the training set only. Tree-based models are invariant to monotonic scaling (Hastie et al., 2009), so scaling does not affect them; it was retained because the Isolation Forest is distance-based and benefits from it, and because Logistic Regression is scale-sensitive.
3. **Splitting** — the initial procedure used a stratified 80/20 train/test split (`random_state = 42`). This split is the one whose results are reported in Section 4.1; Section 4.2 re-evaluates under a leak-proof protocol and explains why the two differ.

3.2 Models

Five models were implemented, all with scikit-learn (Pedregosa et al., 2011):

1. **Baseline threshold rule** — `Fire Alarm = 1 if Temperature[C] > T AND TVOC[ppb] > V`. The thresholds ($T = 15$, $V = 200$) were selected by a coarse grid search maximizing F1 on the training set. This is the interpretable, "rules-first" reference point.
2. **Random Forest** (Breiman, 2001) — 200 trees, `max_depth = 15`, `min_samples_leaf = 5`, `min_samples_split = 10`. Chosen for robustness to multicollinearity among the particulate features (PM1.0/PM2.5 and the NC* counts are strongly correlated) and for its feature-importance output.
3. **Gradient Boosting** (Friedman, 2001) — scikit-learn's `GradientBoostingClassifier` with default hyperparameters. Boosting typically performs well on tabular data (Chen & Guestrin, 2016).
4. **Logistic Regression** — a linear, scale-sensitive baseline.
5. **Isolation Forest** (Liu, Ting, & Zhou, 2008) — an *unsupervised* anomaly detector trained on the full scaled training set with `contamination = 0.05`, included to test whether alarms can be detected as statistical outliers without using the label.

3.3 Evaluation metrics

Because the classes are moderately imbalanced and the operational cost of a false alarm (a needless dispatch) differs from the cost of a missed fire, **precision, recall, and F1 are the primary metrics**; accuracy is reported but not treated as decisive (Powers, 2020). For the leak-proof re-evaluation (Section 4.2), where each held-out recording session is mixed (contains both alarm and no-alarm rows), we additionally report per-session **sensitivity** (recall on the session's alarm rows) and **specificity** (the true-negative rate on the session's no-alarm rows) — the two regime metrics that map directly onto the false-alarm-reduction goal (Fawcett, 2006).

4. Results

4.1 Initial procedure (stratified random split)

Under the stratified 80/20 split, the supervised tree models achieve near-perfect scores on the held-out test set:

Model	Accuracy	Precision	Recall	F1
Random Forest	0.9997	0.9997	0.9999	0.9998
Gradient Boosting	0.9998	0.9998	0.9999	0.9998
Logistic Regression	0.9268	0.9268	0.9745	0.9501
Baseline rule	0.6390	0.8962	0.5596	0.6890
Isolation Forest (unsupervised)	0.2684	0.3450	0.0265	0.0492

Five-fold cross-validation F1 on the full dataset (regular `KFold`) was 0.9395 (± 0.0426) for Random Forest, 0.8697 (± 0.1180) for Gradient Boosting, and 0.7319 (± 0.2689) for Logistic Regression.

Taken at face value, the tree models appear to solve the problem. Section 4.2 shows this face value is misleading.

4.2 Re-evaluation under a leak-proof protocol ("A second look")

A near-perfect F1 on a real-world IoT dataset is a red flag rather than a triumph. The notebook's audit confirms the data is **five continuous recording sessions** separated by gaps of more than sixty seconds, two of which (sessions 1 and 2) are the **same fire recorded twice** — identical row count, duration, and every gas/particulate feature mean; sorted-TVOC correlation = 1.0000. A random train/test split interleaves adjacent one-second-apart rows across the boundary; because adjacent rows are near-duplicate readings that almost always share the same label, a tree model trained on row t effectively sees the answer for row $t + 1$ in the test set. This is a textbook instance of autocorrelation-induced data leakage (Roberts et al., 2017).

After collapsing the duplicate sessions into a single cross-validation unit, there are **four independent sessions**. No leak-free split preserves the 71/29 class balance (the duplicated fire is $\sim 80\%$ of all rows and is entirely alarms), so the evaluation uses **Leave-One-Session-Out (LOSO) cross-validation**: each fold trains on three sessions and is scored on the one held-out session, reporting per-session sensitivity and specificity.

Model	Session	Type	Sensitivity	Specificity	F1
Baseline rule	1	fire (1 + 2)	0.566	0.999	0.723
Baseline rule	3	short fire	0.426	1.000	0.597
Baseline rule	0, 4	quiet	0.000	0.667 / 0.857	0.000
Random Forest	1	fire (1 + 2)	0.000	1.000	0.000
Random Forest	3	short fire	0.000	1.000	0.000
Random Forest	0	quiet	0.000	0.903	0.000
Random Forest	4	quiet	1.000	0.304	0.001
Gradient Boosting	1	fire (1 + 2)	0.264	0.998	0.417
Gradient Boosting	3	short fire	0.000	1.000	0.000
Gradient Boosting	4	quiet	1.000	0.000	0.001

Logistic Regression	1	fire (1 + 2)	0.000	1.000	0.000
Logistic Regression	3	short fire	0.121	0.970	0.216
Isolation Forest	1	fire (1 + 2)	0.000	1.000	0.000
Isolation Forest	3	short fire	1.000	0.000	0.986

(Selected rows shown; the full table is the `loso_results` dataframe in the notebook.)

The picture inverts completely. **Random Forest scores 0.000 sensitivity on every held-out fire session** — it misses 100% of alarms in sessions it has not seen — while staying correctly quiet on one quiet session. **Gradient Boosting** is the only supervised ML model with non-trivial cross-session transfer (sensitivity 0.264 on the held-out fire unit), but it fires on nearly every row of the other held-out quiet session (specificity 0.000). The **Isolation Forest** is bimodal and operationally useless: it either misses all alarms or fires on everything.

Crucially, the **simple threshold rule transfers best** — on the held-out fire session it reaches sensitivity 0.566 with specificity 0.999, catching more than half the real alarms while staying quiet.

5. Interpretation for a Non-Technical Audience

Imagine a smoke detector that, when tested on data it has already seen, appears to be right 99.98% of the time. That sounds perfect — until you ask how it does on a recording it has *never* seen. When we tested that way, the most sophisticated model we built (Random Forest) **missed every single real fire** in the new recording. It had not learned what a fire looks like; it had memorized the specific recording it was trained on.

Why did this happen? The dataset is not 62,000 separate, independent measurements. It is one sensor device recording continuously, one reading per second, for a handful of long sessions. Two of those sessions are actually the same fire recorded twice. When the data is split randomly, a reading from second 1000 lands in the "training" pile and the almost-identical reading from second 1001 lands in the "test" pile — so the model is effectively being handed the answer during evaluation. Its 99.98% score measures "can it recognize this specific recording," not "can it recognize a fire."

The lesson is practical: **a high score is not the same as a working model**. The honest test — train on some recordings, evaluate on a completely different one — revealed that the fancy models fail, while a very simple rule ("call it a fire if it is warmer than 15°C *and* the air has more than 200 parts per billion of volatile compounds") actually holds up reasonably well on unseen data, catching more than half of real fires while rarely crying wolf.

For a real deployment, the takeaway is: don't trust a number until you have checked whether the test data is genuinely separate from the training data. And sometimes the simple, explainable rule is the safer thing to ship.

6. Limitations and Potential Bias

6.1 Limitations and tradeoffs

- **The binding constraint is data diversity, not the algorithm.** The dataset is effectively four independent recording sessions from a single device. The supervised ML models overfit to session-specific sensor signatures and do not generalize across sessions (Section 4.2). No amount of hyperparameter tuning on this data will produce a

model that transfers to a new building; a larger, multi-session, multi-device dataset is required for that.

- **The initial 0.9998 F1 was a leakage artifact.** A random split on autocorrelated data inflates model scores (Roberts et al., 2017); the honest cross-session numbers are roughly four orders of magnitude lower for the tree models. This is both a limitation of the dataset's structure and a demonstration of why a single-number evaluation is insufficient.
- **The threshold rule was tuned on the full dataset.** Its thresholds ($T = 15, V = 200$) reflect this particular prototype and would need per-site re-calibration in a different environment.

6.2 Potential sources of bias

- **Inherited label bias.** The `Fire Alarm = 1` label was assigned by the prototype hardware itself, not by independent ground truth. Any model trained on this data inherits the device's definition of a "fire signature," which may not match a human or regulatory definition. This is a form of label bias that cannot be removed from the data alone.
- **Single-device, single-environment sampling bias.** All recordings come from one prototype in one location; the model has seen no variation in sensor placement, ambient conditions, building type, or occupant behavior. A model deployed elsewhere would face distribution shift it has never encountered (Mehrabi, Morstatter, Saxena, Lerman, & Galstyan, 2021).
- **Class-imbalance artifact.** The 71.5% alarm rate is itself an artifact of one seven-hour fire (recorded twice) dominating the dataset, not a reflection of the true base rate of fires in deployment.

6.3 Steps taken or proposed to reduce these risks

- **Leak-proof evaluation.** We replaced the random split with Leave-One-Session-Out cross-validation that respects the recording session as the independence unit, so the reported numbers reflect genuine cross-session generalization rather than in-session memorization.
- **Honest reporting over vanity metrics.** The near-perfect initial score was preserved in the notebook *and* corrected, rather than silently replaced, so the failure mode is visible and auditable.
- **Proposed for deployment (not yet implemented):** use the interpretable threshold rule as a cold-start detector for new sites, train a per-site ML model only once that site has accumulated its own history, and route low-confidence or distribution-shifted cases to human review. Collecting a multi-device, multi-environment dataset is the single highest-leverage mitigation for the sampling and label biases above.

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